Climate change

a global challenge and the solutions that give us hope

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ENVS 1301

Introduction

Climate change is one of the **major environmental challenges of our time**

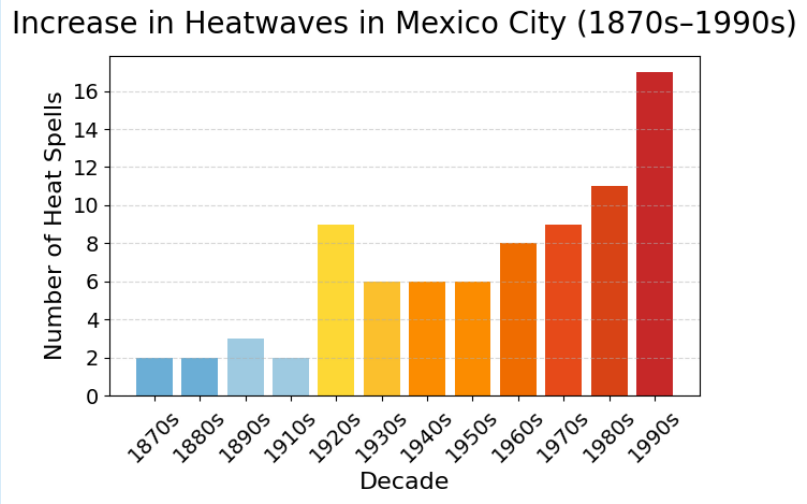
Human activities are increasing **greenhouse gas emissions**, affecting climate and ecosystems

Cities like **Mexico City illustrate both the impacts and the solutions**

This presentation explores:

1. Climate change impacts and solutions
2. Air pollution causes and health effects
3. Waste management and sustainability strategies

Climate change: impact



Long-term alteration of Earth's climate (temperature, precipitation, wind), largely driven by human activities

- **Water cycle**: floods and droughts
- **Food production**: higher temperatures, water changes
- **Human health**: heat-related health issues (heatstroke), infectious diseases (mosquitoes)
- **Environment**: hurricanes, wildfires
- **Infrastructure**: not designed to withstand sea level rise or extreme weather

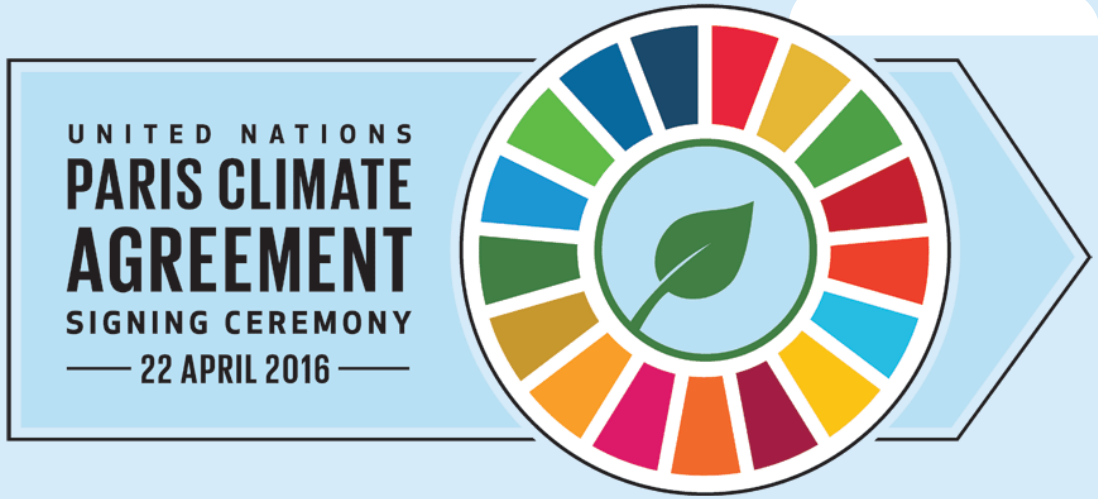
In Mexico:
Higher number of heatwaves

National Oceanic and Atmospheric Administration (2025); Jáuregui (2009)

Climate change: solutions

Global efforts:

- Paris Agreement (194 countries): reduce greenhouse gas emissions to limit the increase of global average temperature to 1.5°C



In **Mexico**:

- Sustainable mobility (⬇ transportation emissions)
- Solar energy (replace fossil fuels and ⬇ emissions)
- Waste management (recycle materials and ⬇ emissions)
- Improve air quality (⬇ pollution and health risks)

United Nations Framework Convention on Climate Change (2026); Secretaría del Medio Ambiente (2021)

Air pollution: what is it and where it comes from?

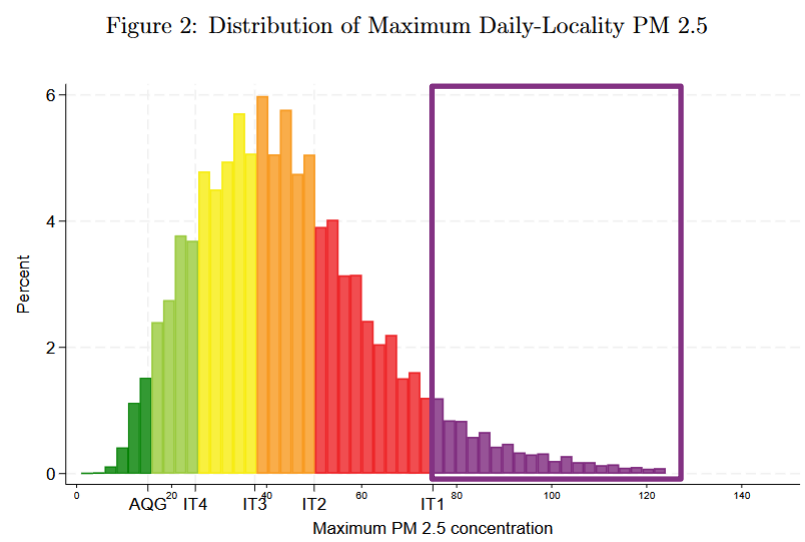
Air pollution: Emission of gaseous and particulate contaminants to the Earth's atmosphere

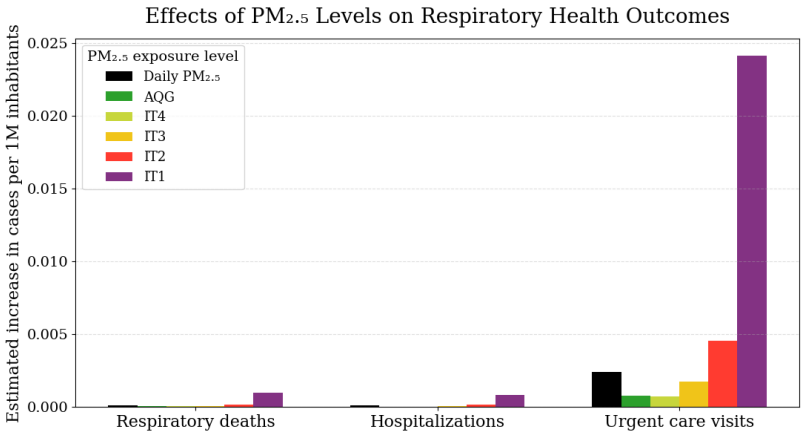
In **Mexico City**:

- Particulate matter (PST, PM_{10}, $\boldsymbol{PM_{2.5}}$):
  - Motor vehicles and industrial facilities
  - Worsened by colder temperatures and lack of rain
- Ground-level ozone (O_3):
  - Motor vehicles and industrial facilities
  - Worsened by higher temperatures and lack of winds
- Sulfur dioxide (SO_2):
  - Industrial facilities northeast of the city (Tula-Tepeji):
    - oil refinery
    - thermoelectric power plant

Harris (2023); René (2016); Secretaría del Medio Ambiente de la Ciudad de México (2020)

Air pollution: what are its consequences?





- Particulate matter ($\boldsymbol{PM_{2.5}}$):
  - Responsible 91% of air quality index peaks
  - Small particles enter lungs and bloodstream: wide-ranging health problems

  - 2025 study (*Inter-American Development Bank*):
    - ⬆ $PM_{2.5}$ levels → ⬆ respiratory deaths, hospitalizations and urgent care visits
    - Short pollution peaks are more harmful than a high daily averages (**black**).

Breathing very polluted air (**purple**) for a short time is more dangerous than moderate pollution over an entire day (**black**).

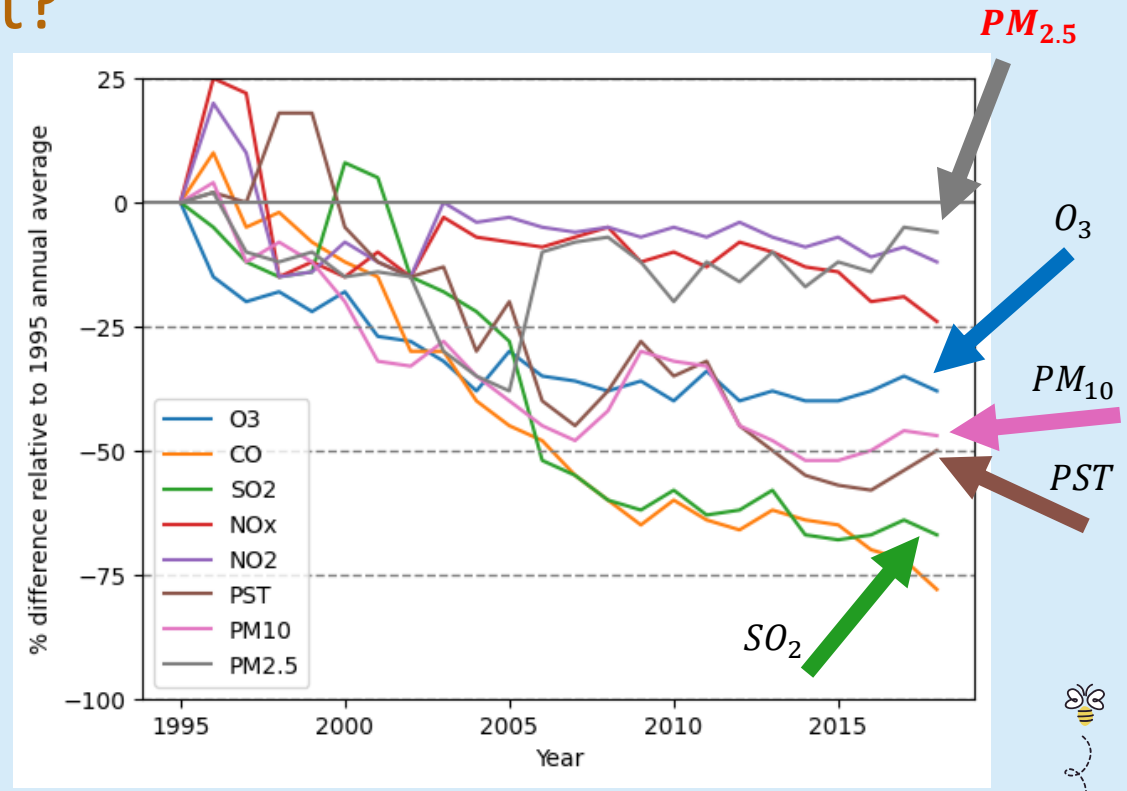
Hoffmann et al. (2025)

Air pollution: what can we do about it?

With more than 20 million people, Mexico City needed to take major steps to improve air quality.

Mitigation strategies:

- Restrict use of vehicles (“Hoy No Circula” program)
- Monitoring and regulating industrial emissions
- Relocating a major refinery



Results: lower levels of particulate matter, ground-level ozone and sulfur dioxide!

Cárdenas et al. (2021); René (2016); Secretaría del Medio Ambiente de la Ciudad de México (2020); UN Environment Programme (2025)

Why waste management matters?

Mexico City: Second biggest producer of waste (NYC is 1st) with 20,000 tons of garbage **every day**!

Lack of waste management:

- landfills and open-air dumping → disease vectors and hazardous gases (air pollution)
- open-air incinerators → air pollution

In fact, waste emissions account for almost 7.4% of total greenhouse gas emissions!

Abubakar et al. (2022); Flores et al. (2023); Kangasmäki (2019)

Waste management: how to make it sustainable?

Strategies

- Separating trash correctly: <u>converting trash into food!</u>
- Circular economy: recycling and reusing (using incentives)
- Landfill gas recovery to supply clean energy

Benefits

- Job creation
- Greenhouse gas and air pollutant mitigation
- Revenue from natural gas and electricity

breathelife2030.org (2018); CNN World (2012); Flores et al. (2023); Gómez (2023); United States Environmental Protection Agency (2025)

Climate change: myths and truths

The Earth's climate is always changing

MYTH!

Truth is:

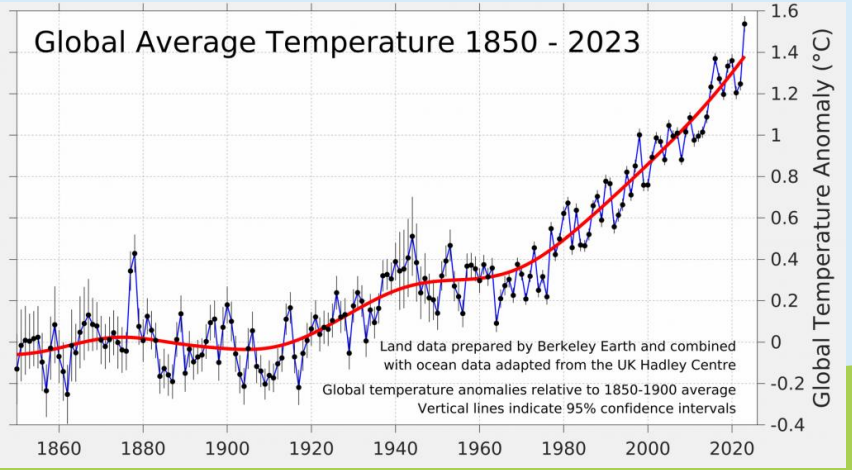
Yes, climate has changed a lot in the 4.5-billion-year history of our planet, **BUT** the rapid warming is **NOT** explained by natural cycles. We are changing our planet!

Truth is:

Global warming increases **average** surface temperature, and this makes weather events like extreme cold more likely. Also, weather is not the same as climate: we can have cold winters, but our winters now are warmer than 100 years ago!

MYTH!

Global warming isn't real because it is still cold



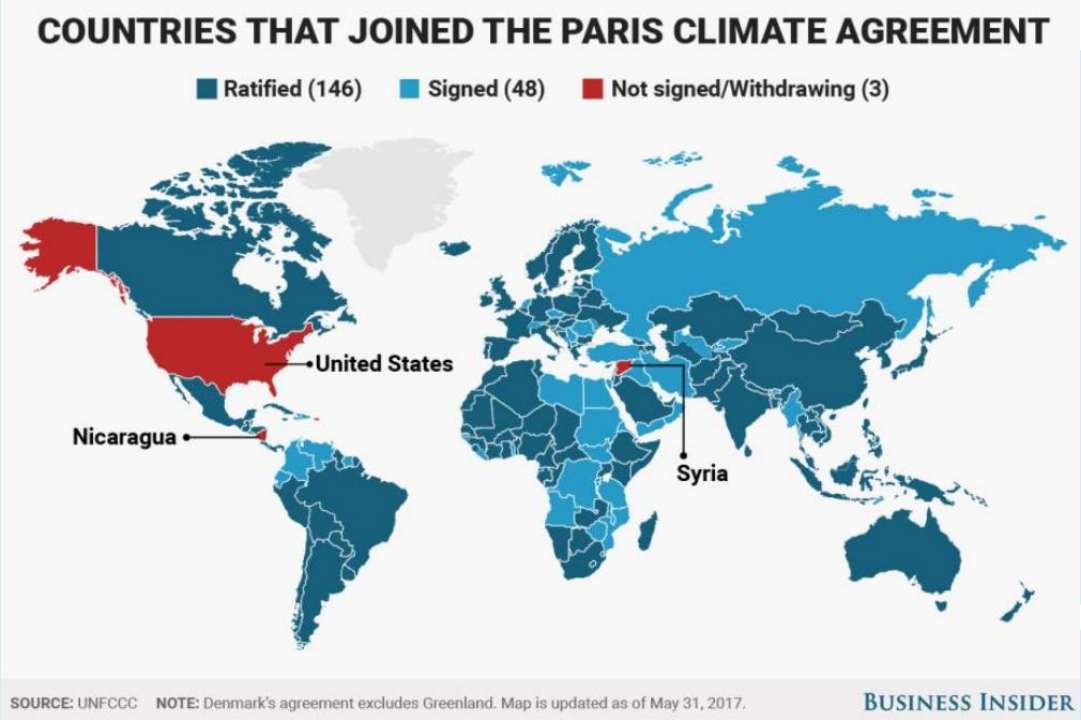
World Wide Fund for Nature (n.d.); Rohde (2024)

Climate change: myths and truths

People believe disinformation about climate change

Public pressure on climate action policies decreases

Political decisions weaken or delay environmental policies



United Nations Economic Comission for Africa (n.d.)

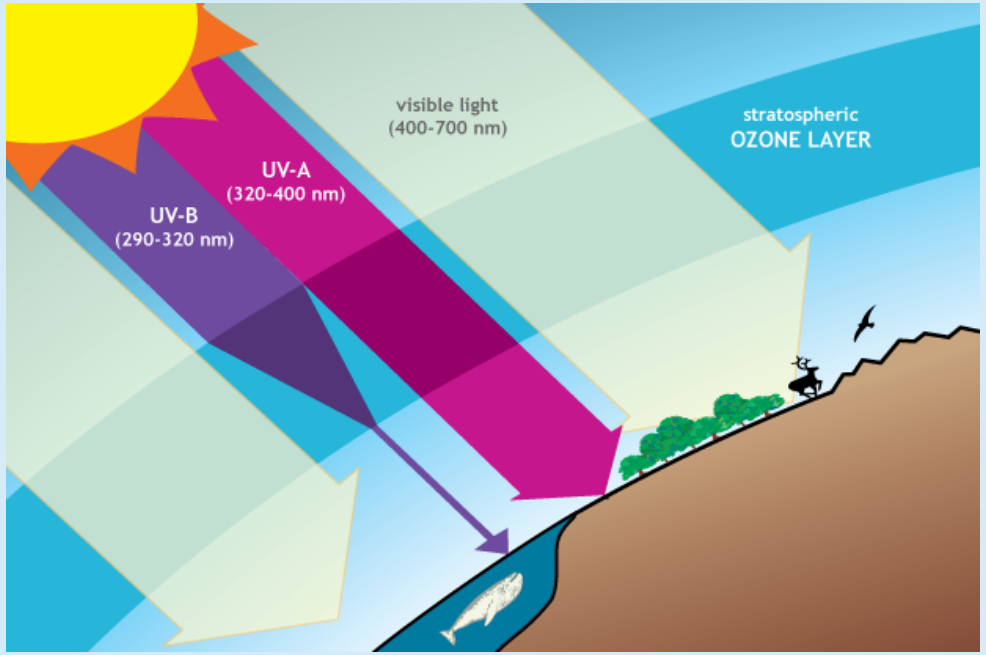
The Ozone layer: why is it important?

Role of Ozone layer

- Absorbs harmful ultraviolet (UV) radiation form the Sun
- Ozone-depleting substances (CFCs) are greenhouse gases!
- And therefore, protecting Ozone = reduce global warming

Notes

- In 1974, Mario Molina (Nobel Prize - Mexico) and colleagues showed that CFC gases destroy ozone in the stratosphere
- The **Montreal Protocol (1987)** phased out these chemicals worldwide.



Harris (2023); Lindsey (2016); Montreal Protocol Ozone Secretariat (2025); NobelPrize.org (n.d.)

Conclusion

Climate change affects **ecosystems, human health, food production, and infrastructure**

Air pollution and waste management are closely linked to climate change

Policies in Mexico City show that **environmental action can improve air quality**

Global cooperation, such as the **Paris Agreement and Montreal Protocol**, demonstrates that solutions are possible

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